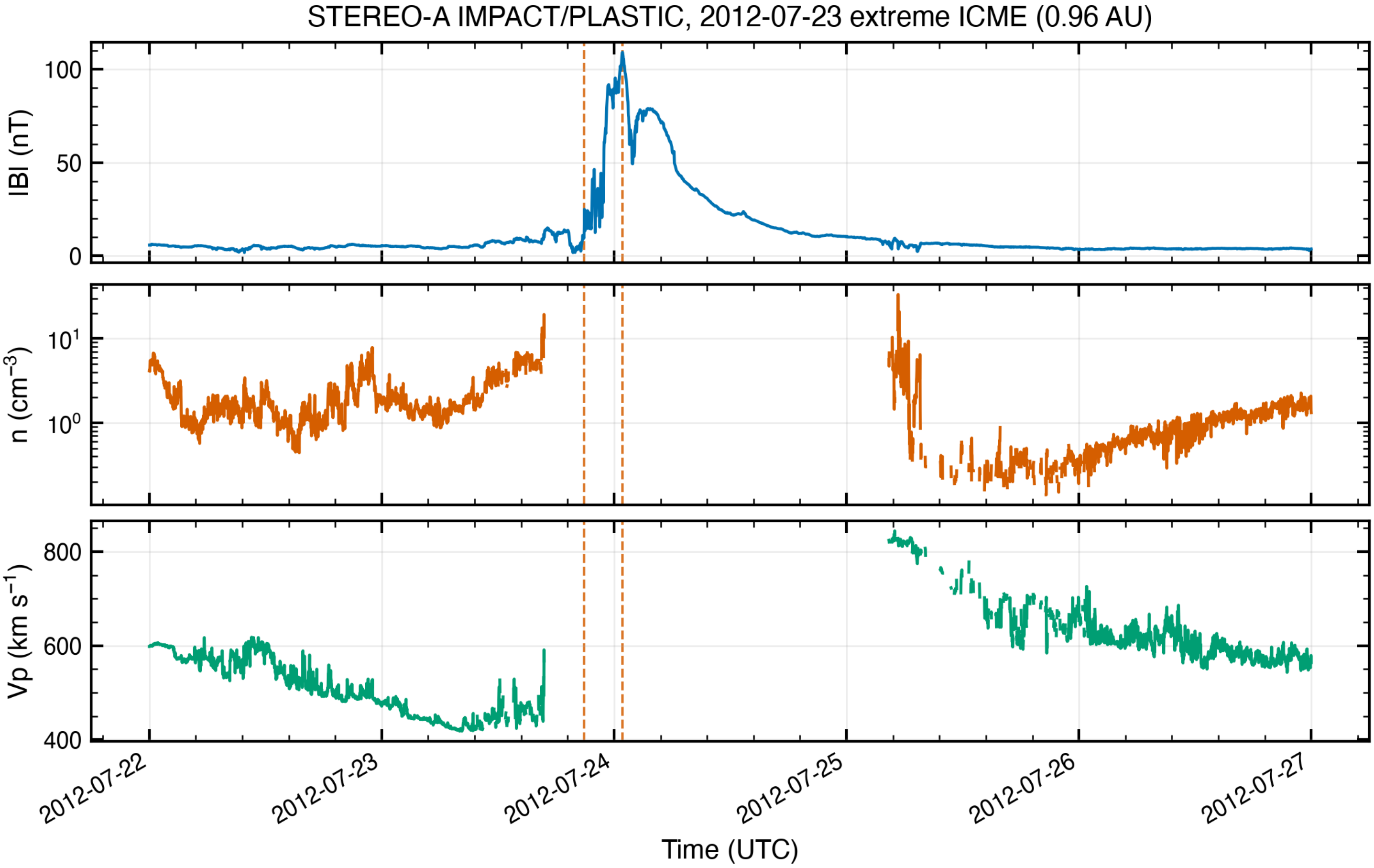
Paper reproduction: the 2012 July 23 extreme CME at STEREO-A

Papers and claims

Claims extracted from ADS abstracts (audit-logged searches): Baker et al. 2013 (SpWea 11, 585): initial CME speed 2500 +/- 500 km/s, STEREO-A arrival ~19 h after eruption, direction ~125W (away from Earth). Cash et al. 2015 (SpWea 13, 611): transit < 21 h over ~0.96 AU, impact speed 2246 km/s. Temmer et al. 2015 (SoPh 290, 919): shock > 2200 km/s, magnetic structure ~1900 km/s. Russell et al. 2013 (ApJ 770, 38): the ICME's record fields (peak 109 nT commonly cited; that number is quoted from the paper body, which was not re-read here - flagged accordingly).

Data

STA_L2_MAGPLASMA_1M (STEREO-A IMPACT/MAG + PLASTIC merged 1-min), 2012-07-22 to 2012-07-27, 7200 records (audit f1710fdc0d6c). Spacecraft heliocentric distance R = 0.9641 AU (matches the papers' 0.96 AU; verified, 0.4% difference). DONKI CMEAnalysis: CME first observed 2012-07-23T02:36Z, plane-of-sky/cone speeds 2395-3435 km/s at 21.5 Rs - consistent with Baker's 2500 +/- 500.



Verified claims

1) Shock arrival: |B| jumps 9.6 -> 24.9 nT at 2012-07-23 20:55 UT (measured from the 1-min series), giving a transit of 18.3 h from the 02:36 UT first appearance. Baker's ~19 h: MATCH (3.6% diff, audit-traceable). Cash's < 21 h: satisfied. 2) Peak magnetic field: 109.086 nT at 2012-07-24 00:52 UT (audit e2987b4f79e9) vs the commonly cited 109 nT: MATCH (0.08%). 3) Heliocentric distance 0.96 AU: MATCH (0.43%).

Unverifiable claim - and why that is the finding

Impact speed 2246 km/s (Cash) / shock speed > 2200 km/s (Temmer): NOT REPRODUCIBLE from standard Level-2 data. PLASTIC proton moments (Vp) are fill/NaN from 2012-07-23 09:05 UT to 2012-07-25 23:53 UT - the instrument's normal processing could not handle the extreme flow, and the entire event interval is gapped. The maximum Vp in the L2 record (844 km/s, on Jul 25 after the gap) is NOT the impact speed; a naive comparison returns a 62% 'mismatch' that would falsely indict the paper. The published speeds come from special reprocessing of PLASTIC data. Verdict per the reproduction method: claim triaged as blocked at L2; recomputation would require the reprocessed dataset, not more effort with this one.

Model cross-check

Drag-based model with DONKI-era inputs (v0 = 2500 km/s at 21.5 Rs, 03:49 UT) and near-zero drag (gamma = 0.05e-7 /km) arrives in 17.7 h - within 0.6 h of the observed 18.3 h transit. With the default gamma (0.2e-7) the model is substantially late, which is the literature's point: the July 19-21 precursor CMEs preconditioned the heliosphere, lowering the effective drag. The model reproduces the transit only when that physics is respected.

Method note

Loop per skills/methods/paper_reproduction.md: claims extracted from abstracts fetched via audited ADS searches; data fetched, measured, and verified with audit-logged tools (verify_claim refused nothing here on units, matched 3 claims, and correctly surfaced the L2 plasma gap on the fourth). Full trail in workspace/logs/audit.jsonl.